

1.7.14: Questions

Quick questions

You should be able to answer these questions without too much difficulty after studying this TLP. If not, then you should go through it again!

Which of these is **not** a lattice type?

- | | |
|---|---|
| a | H |
| b | F |
| c | I |
| d | C |

Answer

A.

What is a lattice ?

- | | |
|---|---|
| a | A transforming matrix |
| b | A set of atoms |
| c | A group of symmetry elements |
| d | An infinite array of identical points repeated throughout space |

Answer

D.

How many Bravais lattices (3D) are there ?

a	36
b	28
c	14
d	16

Answer

C.

How many point groups (3D) are there ?

a	24
b	32
c	12
d	19

Answer

B.

What is a unit cell ?

a	A unit of volume
b	Any parallelepiped with lattice points at its corners
c	A parallelepiped containing only one lattice point
d	The angle between lattice vectors

Answer

B.

Deeper questions

The following questions require some thought and reaching the answer may require you to think beyond the contents of this TLP.

Construct a plan view of NaCl (sodium chloride).

NaCl has a **face-centred cubic** lattice. The motif is:

Cl @ (0,0,0);

Na @ (0,0,1/2);

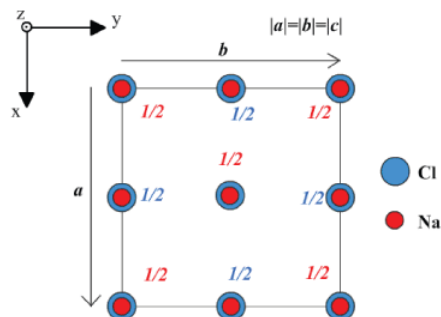
Note 1: The motif coordinates are positions relative to each lattice point

Note 2: In a face centred cubic structure the lattice points are located at:

(0,0,0), (1/2,1/2,0), (1/2,0,1/2), (0,1/2,1/2)

Answer

Your construction should look something like this



It should include:

A set of axes

The **a** and **b** lattice vectors

Two separate symbols for Cl and Na plus a key explaining the symbols

The fractional heights of the atoms.

Diamond has a **face centred cubic** lattice. Its motif is

C @ (0,0,0), (1/4,1/4,1/4)

Construct a plan view of the diamond unit cell.

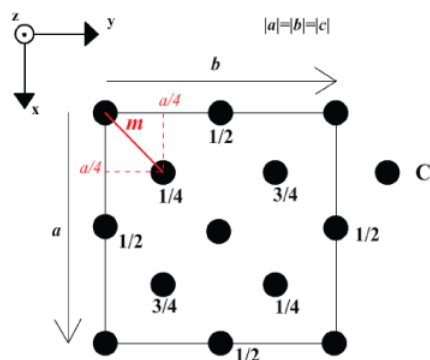
Treating the carbon atoms as hard spheres calculate the packing efficiency of diamond.

Note: In a face centred cubic structure the lattice points are located at:

(0,0,0), (1/2,1/2,0), (1/2,0,1/2), (0,1/2,1/2)

Answer

Your construction should look something like this



Packing Efficiency:

The volume of the unit cell is a^3 .

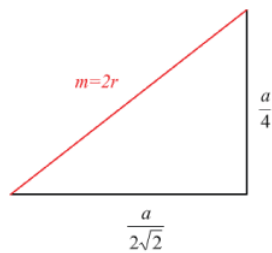
The volume of a single atom is

$$\frac{4}{3}\pi r^3 \quad (1.7.14.1)$$

There are 8 atoms per unit cell.

Note that the solid red line extends to a height of $a/4$. The total length of m is equal to $2r$ (the centre-centre distance of two touching atoms is $2r$). The distance covered by m in the xy plane is .

$$\frac{a}{2\sqrt{2}} \quad (1.7.14.2)$$



We can see from this diagram that

$$2r = \frac{\sqrt{3}}{4}a \quad (1.7.14.3)$$

therefore

$$r = \frac{\sqrt{3}}{8}a \quad (1.7.14.4)$$

The volume of a single atom is

$$\frac{4}{3}\pi \left(\frac{\sqrt{3}}{8}a \right)^3 = \frac{\sqrt{3}}{128}\pi a^3 \quad (1.7.14.5)$$

The volume occupied by the 8 atoms is

$$\frac{\sqrt{3}}{16}\pi a^3 \quad (1.7.14.6)$$

Therefore the packing efficiency is

$$\frac{\sqrt{3}}{16}\pi a^3 \div a^3 \approx 0.34 \quad (1.7.14.7)$$

$$\Rightarrow 34\% \quad (1.7.14.8)$$

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